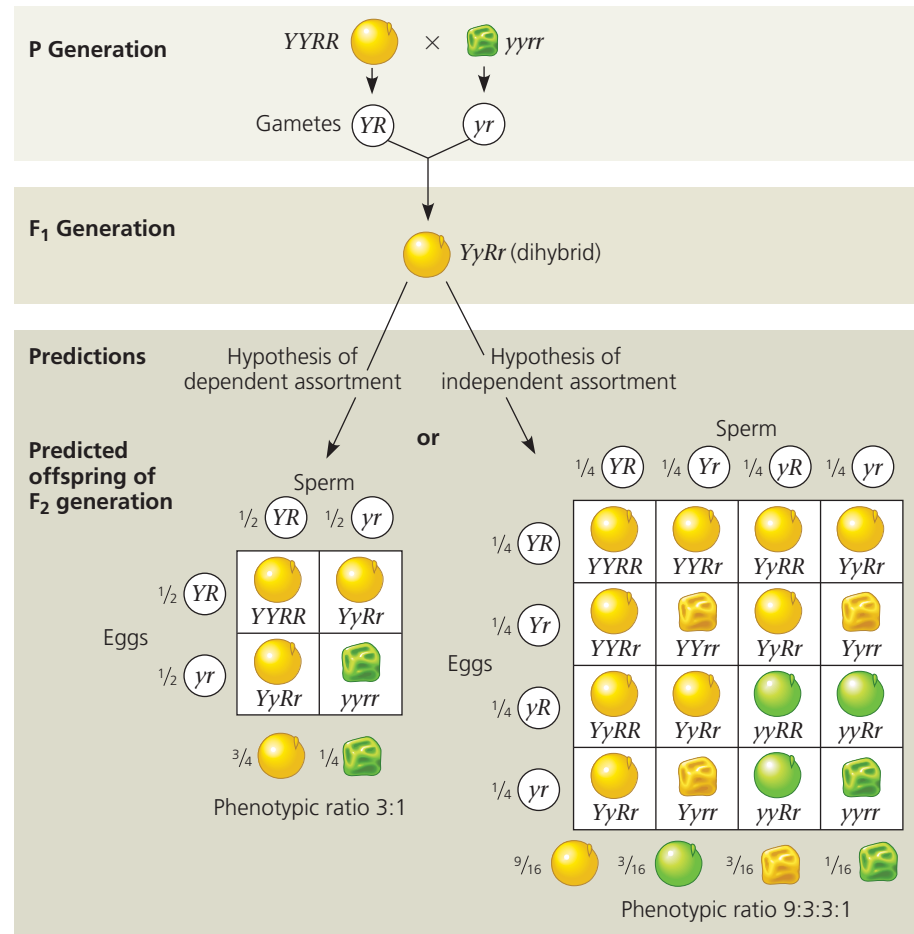


## ▼ Figure 14.8 Inquiry

**Do the alleles for one character segregate into gametes dependently or independently of the alleles for a different character?**

**Experiment** To follow the characters of seed color and seed shape through the  $F_2$  generation, Mendel crossed a true-breeding plant with yellow round seeds with a true-breeding plant with green wrinkled seeds, producing dihybrid  $F_1$  plants. Self-pollination of the  $F_1$  dihybrids produced the  $F_2$  generation. The two hypotheses (dependent and independent “assortment” of the two genes) predict different phenotypic ratios.



## Results

315 yellow round, 108 green round, 101 yellow wrinkled, 32 green wrinkled. Phenotypic ratio approximately 9:3:3:1

**Conclusion** The results support the hypothesis of independent assortment, the only one that predicts two newly observed phenotypes: green round seeds and yellow wrinkled seeds (see the right-hand Punnett square). The alleles for each gene segregate independently of those of the other, and the two genes are said to assort independently.

**Data from** G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brunn* 4:3–47 (1866).

**WHAT IF?** Suppose Mendel had transferred pollen from an  $F_1$  plant to the carpel of a plant that was homozygous recessive for both genes. Set up the cross and draw Punnett squares that predict the offspring for both hypotheses. Would this cross have supported the hypothesis of independent assortment equally well?

## ➔ Mastering Biology

**Animation:** Mendel's Cross of Two Characters: Seed Shape and Seed Color  
**Animation:** A Simplified Cross of Two Characters in Humans  
**Animation:** Crosses of Two Characters in “MendAliens”

## CONCEPT CHECK 14.1

- DRAW IT** Pea plants heterozygous for flower position and stem length ( $AaTt$ ) are allowed to self-pollinate, and 400 of the resulting seeds are planted. Draw a Punnett square for this cross. How many offspring would be predicted to have terminal flowers and be dwarf? (See Table 14.1.)
- WHAT IF?** List all gametes that could be made by a pea plant heterozygous for seed color, seed shape, and pod shape ( $YyRrIi$ ; see Table 14.1). How large a Punnett square would you need to draw to predict the offspring of a self-pollination of this “trihybrid”?
- MAKE CONNECTIONS** In some pea plant crosses, the plants are self-pollinated. Is self-pollination considered asexual or sexual reproduction? Explain. (See Concept 13.1.)

For suggested answers, see Appendix A.

## CONCEPT 14.2

## Probability laws govern Mendelian inheritance

Mendel's laws of segregation and independent assortment reflect the same rules of probability that apply to tossing coins, rolling dice, and drawing cards from a deck. The probability scale ranges from 0 to 1. An event that is certain to occur has a probability of 1, while an event that is certain *not* to occur has a probability of 0. With a coin that has heads on both sides, the probability of tossing heads is 1, and the probability of tossing tails is 0. With a normal coin, the chance of tossing heads is  $\frac{1}{2}$ , and the chance of tossing tails is  $\frac{1}{2}$ . The probability of drawing the ace of spades from a 52-card deck is  $\frac{1}{52}$ . The probabilities of all possible outcomes for an event must add up to 1. With a deck of cards, the chance of picking a card other than the ace of spades is  $\frac{51}{52}$ .

Tossing a coin illustrates an important lesson about probability. For every toss, the probability of heads is  $\frac{1}{2}$ . The outcome of any particular toss is unaffected by what has happened on previous trials. We refer to phenomena such as coin tosses as independent events. Each toss of a coin, whether done sequentially with one coin or simultaneously with many, is independent of every other toss. And like two